

White is an appearance, not a spectrum.

It is a spectral construction that happens to land on a white appearance.

READING RULE

Read where the energy is.

Warm, neutral, and cool describe how the source appears. The spectral power distribution shows how that appearance was actually constructed: where the source rises, where it falls away, and which wavelengths do most of the work.

WHAT THE EYE HIDES

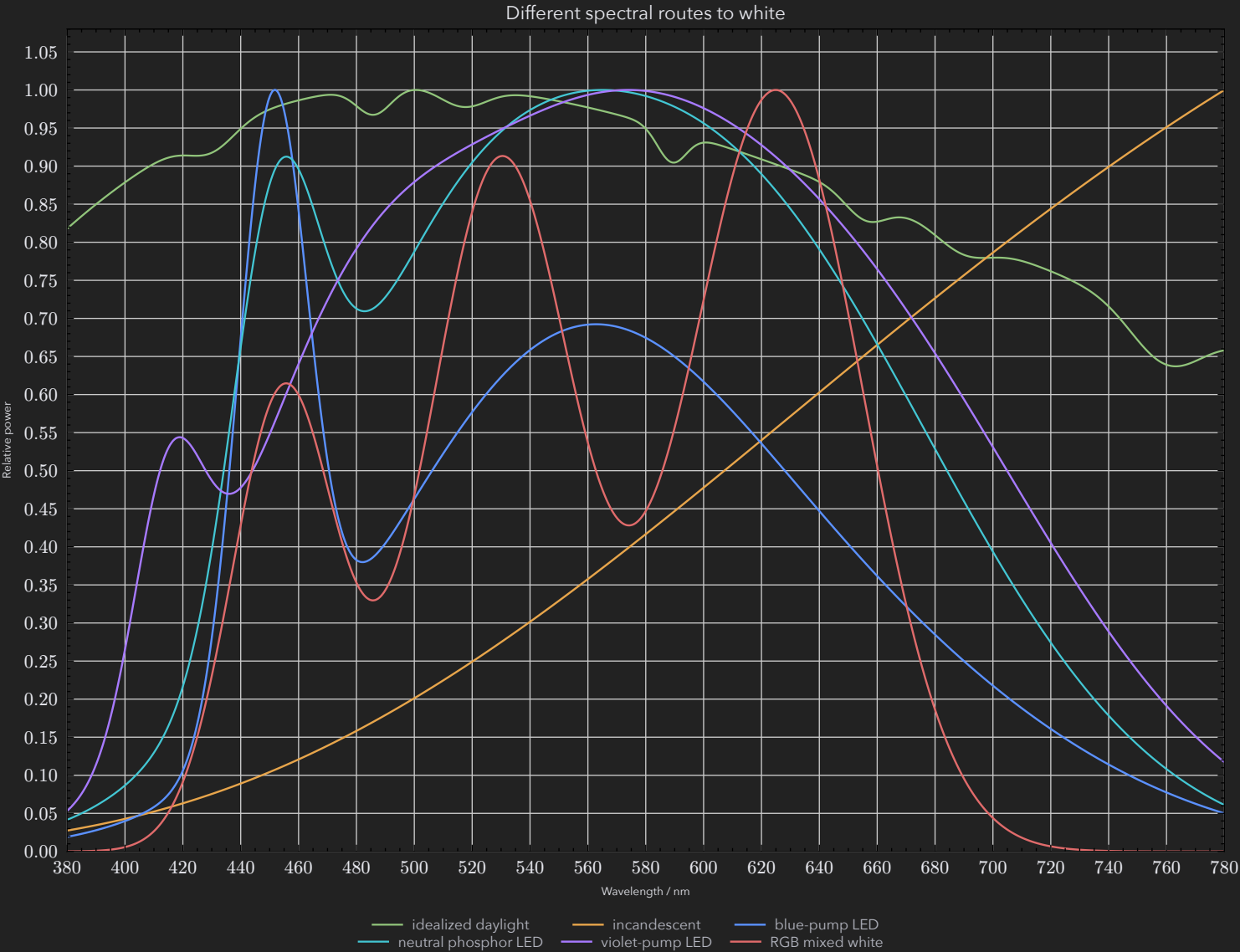
Chromaticity is a compression.

The visual system can collapse very different spectral distributions into a similar judgment of white. Matching appearance therefore does not imply matching spectral structure.

SOURCE VERSUS RESULT

“White” tells you where the source landed – not how it got there.

A glowing filament, a blue LED driving phosphor, a violet pump, or separate red, green, and blue emitters can all arrive at a white chromaticity by very different physical routes.



SAME CATEGORY

All six sources can occupy the broad visual category of white, despite radically different distributions underneath.

DIFFERENT PHYSICS

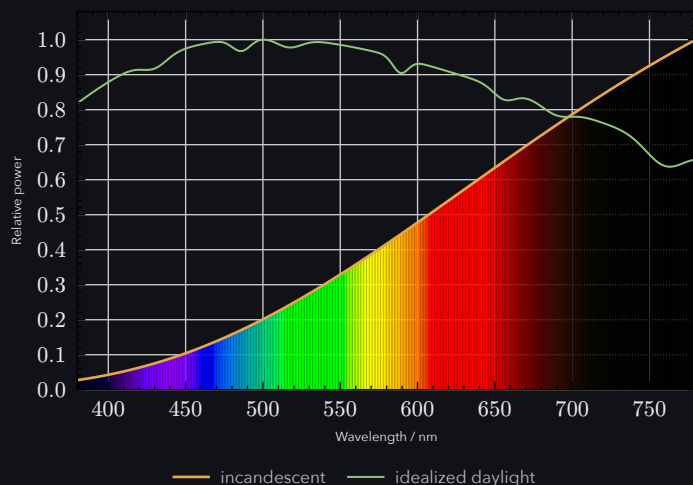
The peaks, slopes, gaps, and continua are fingerprints of the physical mechanism producing the light.

FIRST RULE

If the question concerns what the light physically contains, start with the spectrum. CCT is a useful visual summary; it is not a source description.

THERMAL SOURCE

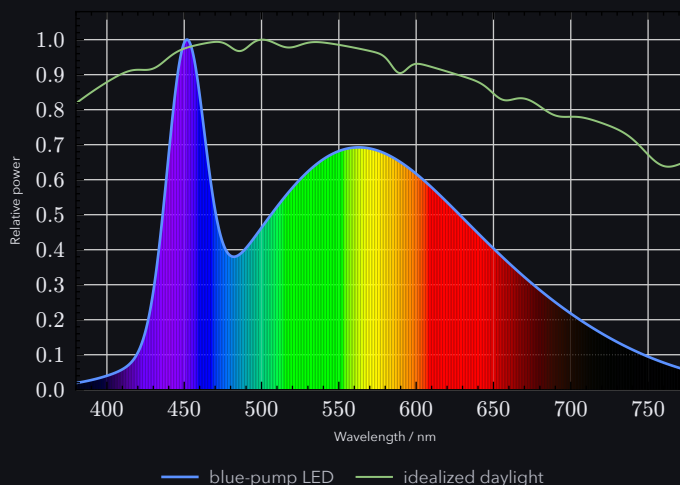
Continuous does not mean daylight-like.



Incandescent output forms a smooth thermal continuum, but the distribution is heavily weighted toward red and longer wavelengths.

PHOSPHOR SOURCE

White appearance can hide a pump spike.



The blue pump remains a distinct spectral feature even after the phosphor contribution broadens the source into visually white light.

SOURCE ARCHITECTURE

How white light is made.

The spectrum reveals the physical process behind the appearance: thermal emission, phosphor conversion, or narrow-band mixing.

THERMAL EMISSION

A continuum can still be strongly biased.

Incandescent light comes from a hot radiator. Its spectrum is smooth, but it climbs strongly toward longer wavelengths and continues beyond the visible range into infrared. Continuous does not mean daylight-like.

PHOSPHOR CONVERSION

A white lamp can retain the pump that created it.

In a blue-pump LED, a narrow blue emitter excites a much broader phosphor band. The resulting mixture can look ordinary and balanced while the pump remains plainly visible in the spectrum.

NARROW-BAND MIXING

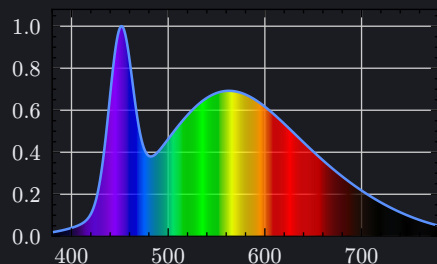
White does not require a continuous spectrum.

Red, green, and blue emitters can be mixed to a white chromaticity even though large portions of the visible range remain comparatively sparse.

THREE ENGINEERED ROUTES

BLUE PUMP

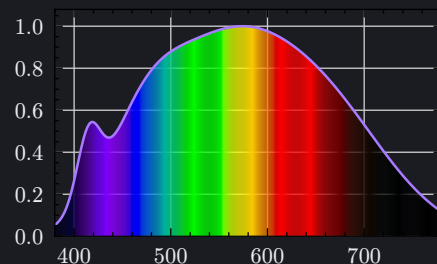
Phosphor white



A narrow blue emitter supplies the pump while a broad converted band fills the middle and long wavelengths.

VIOLET PUMP

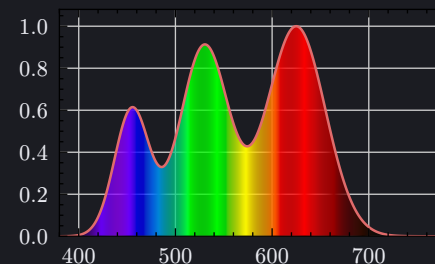
Broader conversion



Moving the pump toward violet allows the converted spectrum to distribute output more broadly across the visible range.

ADDITIVE MIXING

RGB white



Three narrow emitters can converge on white appearance without filling the wavelengths between their individual peaks.

These are not merely different colors of white. They are different source architectures that happen to produce a similar visual category.

USEFUL QUESTION

Do not stop at "what color is this light?" Ask what physical mechanism produced it, and what spectral structure that mechanism leaves behind.

Spectrum is read by more than one system.

The retina does not send one report to the brain. It sends a visual report and a biological timing report.

THE HIDDEN READER

The same photons carry two meanings.

Beyond rods and cones, the retina contains intrinsically photosensitive retinal ganglion cells, or ipRGCs. These cells project through the retinohypothalamic tract to the suprachiasmatic nucleus, the brain's master circadian clock, and to pathways involved in pupil constriction, alertness, morning cortisol release, and melatonin suppression.

VISUAL SYSTEM

Photopic – $V(\lambda)$

Peaks near 555 nm. It is the basis for lux, footcandles, and luminous efficacy – the quantities that tell a lighting designer how bright the room appears.

NON-VISUAL SYSTEM

Melanopic – ipRGC

Peaks near 480–490 nm. It is produced by melanopsin in intrinsically photosensitive retinal ganglion cells and is quantified in melanopic EDI.

READING THE NUMBER

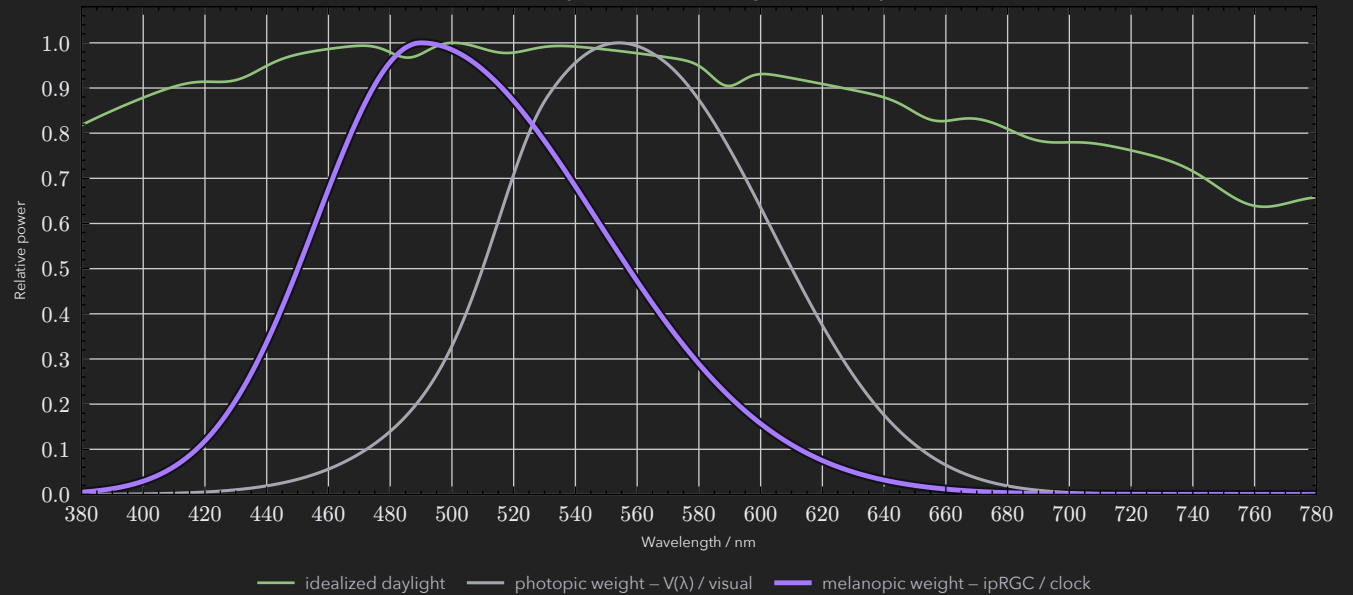
What the pill numbers on the right mean.

Each modeled source is evaluated by melanopic DER: its melanopic response is compared with its photopic response, then referenced to D65 daylight, which has a DER of 1.0. Higher DER means more melanopic stimulus for the same amount of visual light.

FIRST RULE

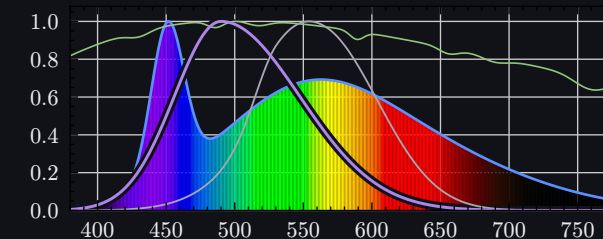
Brightness, warmth, CCT, and CRI describe the visual layer. They do not prove what the source does biologically. The circadian layer requires reading the spectrum itself.

Two sensitivity functions reading the same spectrum



SAME VISUAL TARGET

Cheap blue-pump LED

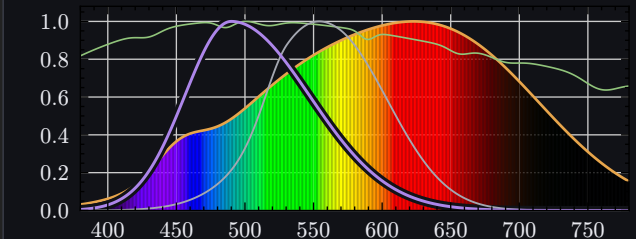


≈ 0.82 melanopic DER

Ordinary white light, but the pump peak sits inside the ipRGC window: a strong clock signal riding on an unremarkable-looking lamp.

SAME VISUAL TARGET

Warm phosphor LED



≈ 0.56 melanopic DER

Comparable brightness, warmer appearance, and a somewhat lower melanopic DER: a modest reduction in clock drive, not an escape from it.

Both lamps can look reasonably white and pass ordinary lux, CCT, and CRI checks – the two curves above are what actually distinguish them biologically.

IMPLEMENTATION TARGETS

Specify the light that reaches the eye.

Circadian lighting is not specified by CCT, fixture wattage, or horizontal lux alone. The practical design quantity is melanopic equivalent daylight illuminance at the observer’s eye, evaluated in time as well as space.

THE DESIGN QUANTITY

Melanopic EDI is the scene target.

CIE S 026 expresses melanopsin-weighted exposure as melanopic equivalent daylight illuminance: the illuminance of standard D65 daylight that would produce the same melanopic stimulus. For implementation, this is the quantity to calculate or measure at the eye.

MEASUREMENT PLANE

Measure vertically, where the eye is looking.

Conventional horizontal workplane lux does not describe retinal exposure. Evaluate melanopic EDI on a vertical plane at representative eye position and facing the expected direction of gaze.

IMPORTANT DISTINCTION

DER tells you the engine. EDI tells you the delivered signal.

Melanopic DER describes the spectral efficiency of a source relative to D65 daylight. Melanopic EDI combines that spectral property with the actual photopic illuminance reaching the eye.

DAYTIME

$\geq 250 \text{ lx}$

melanopic EDI at the eye

Target throughout the normal waking day for healthy day-active adults. Daylight is the preferred way to reach or exceed the target where practical.

EVENING

$\leq 10 \text{ lx}$

during the 3 h before bedtime

Reduce retinal melanopic exposure substantially before sleep. This is a scene-level limit, so displays, windows, decorative light, and task lighting all contribute.

SLEEP

$\leq 1 \text{ lx}$

through the sleep period

Darkness is the design condition during sleep. If nighttime visual activity is necessary, provide only the visibility required while keeping melanopic exposure very low.

HOW THE QUANTITIES FIT TOGETHER

spectral power distribution

what wavelengths the source emits



melanopic DER

melanopic efficiency relative to D65



melanopic EDI

delivered melanopic exposure at the eye

PHOTOPIC ILLUMINANCE

LX

visual quantity

Ordinary illuminance weighted by V(λ). It remains the conventional quantity for visibility, tasks, glare assessment, and ordinary lighting design.

MELANOPIC DER

RATIO

source property

The melanopic daylight efficacy ratio. It indicates how much melanopic stimulus a spectrum produces per unit of photopic illuminance relative to D65. D65 has a DER of 1.

MELANOPIC EDI

LX

scene exposure

The primary circadian implementation quantity: the D65-equivalent illuminance producing the same melanopic stimulus as the actual light reaching the observer’s eye.

WORKING RELATIONSHIP

Melanopic EDI ≈ vertical photopic lux × melanopic DER.

For a source or scene whose spectrum is appropriately characterized:

PHOTOPIC 500 lx × MELANOPIC DER 0.55 = MELANOPIC EDI 275 lx

The same spectral engine at only 20 vertical photopic lux produces approximately 11 lx melanopic EDI. Spectrum determines conversion efficiency; intensity determines how much signal is delivered.

COMMISSIONING

Verify the occupied eye position.

Do not sign off a circadian scene from fixture data alone. Measure or model the complete scene at representative eye locations and gaze directions, including daylight and reflected light.

CONTROL SEQUENCE

Targets belong to time states.

A fixture does not have one universally healthy setting. Commission daytime, evening, and nighttime scenes separately and verify melanopic EDI in each state.

DER connects photopic illuminance to melanopic exposure.

Once the spectrum’s DER is known, the same relationship can answer two opposite design questions: how much light is required by day, and how much can remain in the evening.

ONE EQUATION, TWO QUESTIONS

The arithmetic stays the same. The unknown changes.

For daytime, start with a melanopic target and solve for the photopic illuminance required to reach it. For evening, start with a melanopic ceiling and solve for the photopic illuminance that can remain beneath it.

SAME 100 LX, DIFFERENT SPECTRUM

DER 1.0

100 lx → 100 lx melanopic EDI

A daylight-like ratio: melanopic EDI tracks photopic illuminance one-for-one.

DER 0.5

100 lx → 50 lx melanopic EDI

The same visual-light level carries half as much melanopic exposure.

DER 0.2

100 lx → 20 lx melanopic EDI

Much less melanopic exposure remains for the same photopic illuminance.

THE REVERSAL

The desirable direction depends on time.

Higher DER is useful when the goal is to build melanopic exposure efficiently. Lower DER is useful when the goal is to preserve visual light while constraining melanopic exposure.

FROM VISUAL LIGHT TO MELANOPIC EXPOSURE

photopic lx

×

melanopic DER

melanopic EDI

Change the photopic level and the total exposure changes. Change DER and the melanopic content of each photopic lux changes.

SOLVE FOR THE UNKNOWN

BIOLOGICAL DAY

How much light is required?

Suppose the daytime target is 250 lx melanopic EDI.

$250 \div \text{DER} = \text{required photopic lx}$

At DER 1.0, the target requires 250 photopic lx.

At DER 0.5, it requires 500 photopic lx.

Higher DER therefore reaches the same melanopic target with less photopic illuminance.

BIOLOGICAL EVENING

How much light can remain?

Suppose melanopic EDI must stay at or below 10 lx.

$10 \div \text{DER} = \text{maximum photopic lx}$

At DER 1.0, only 10 photopic lx can remain.

At DER 0.5, 20 photopic lx can remain.

Lower DER therefore preserves more visual light beneath the same melanopic ceiling.

HIGHER DER → MORE EDI PER LX ↔ LOWER DER → LESS EDI PER LX

THE TARGET CHANGES WITH TIME

NORMAL WAKING DAY

≥ 250 lx melanopic EDI

Establish a strong daytime signal.

→

3 H BEFORE BED

≤ 10 lx melanopic EDI

Reduce melanopic exposure while retaining useful light.

→

SLEEP PERIOD

≤ 1 lx melanopic EDI

Keep necessary nighttime exposure tightly constrained.

The equation does not reverse between day and evening. The design objective does: first use the spectrum to build melanopic exposure, then use it to limit that exposure without discarding more visual light than necessary.

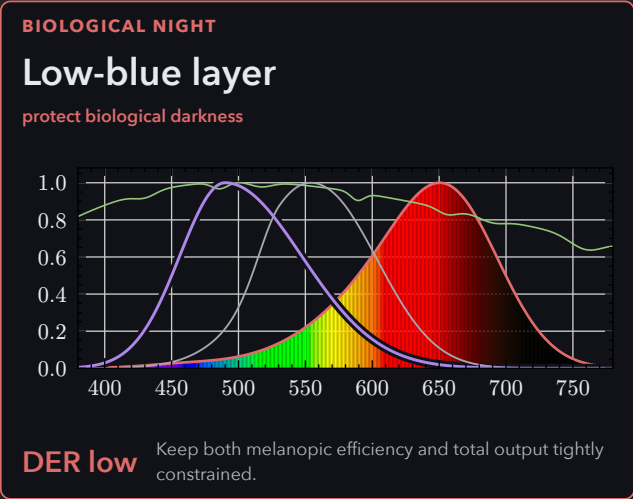
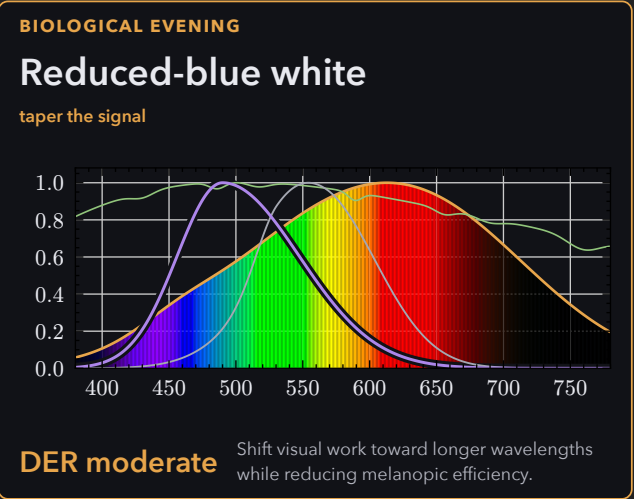
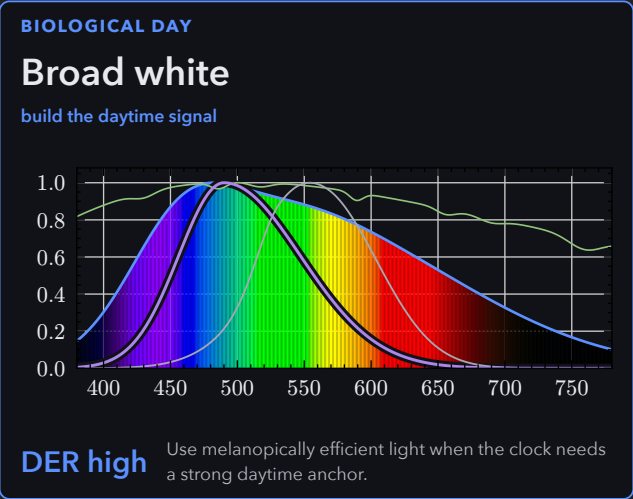
DESIGN RULE

Set the biological target first. Then choose illuminance and spectrum together to reach it.

The target is not one perfect spectrum.

A circadian lighting system should change its spectral engine as the biological objective changes. Day calls for a strong melanopic signal; evening calls for withdrawal; biological night calls for protection.

THREE SPECTRAL STATES



The spectra above are illustrative control states rather than product specifications. Actual DER depends on the emitted spectral power distribution.

IMPORTANT DISTINCTION

Low DER is not the same as low exposure.

DER changes the conversion between visible light and melanopic signal. Illuminance still determines how much light is actually delivered.

20 lx × DER 1.0 = 20 lx EDI

High-DER light can still produce modest melanopic exposure when the illuminance is low.

50 lx × DER 0.2 = 10 lx EDI

Lower DER permits more visible light for the same melanopic budget, but it does not make high output biologically irrelevant.

THREE CONTROL LEVERS

SPECTRUM

DER

Which wavelengths are present, and how efficiently does the spectrum generate melanopic stimulus per unit of visual light?

INTENSITY

Photopic illuminance

How much visible light reaches the eye? DER only becomes exposure when it is multiplied by actual illuminance.

TIME

Schedule

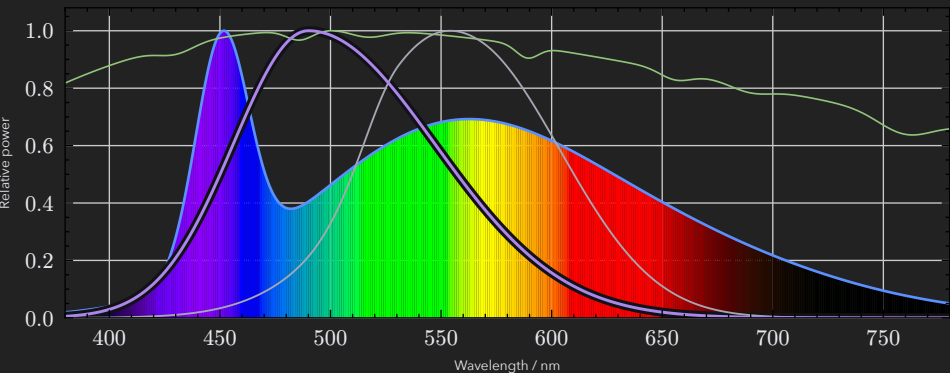
When is the signal delivered? The same melanopic exposure can be appropriate during biological day and disruptive near sleep.

Read the spectrum, not just the label.

A lamp can look warm, score well on color rendering, and still deliver substantial melanopic stimulus. Read a source in sequence: spectrum, melanopic DER, illuminance at the eye, then time of use.

START WITH THE SPECTRUM

The shape contains information the package cannot.



Example: a conventional blue-pump white LED. The visible appearance collapses this spectrum into a single impression of “white,” while the spectral plot exposes the short-wavelength pump and broad phosphor output.

1

SPECTRUM
What wavelengths are emitted?
Look for the spectral power distribution. Similar-looking whites can distribute their energy very differently across the visible range.

2

MELANOPIC DER
How efficient is the spectrum?
DER expresses melanopic response per unit of photopic illumination relative to D65 daylight, whose DER is 1.

3

ILLUMINANCE
How much reaches the eye?
Measure or model vertical photopic illuminance at the observer. Spectrum alone does not determine delivered exposure.

4

TIME
When is the signal delivered?
High melanopic exposure may support biological day while the same exposure near bedtime may oppose the intended transition to night.

WORKING SEQUENCE

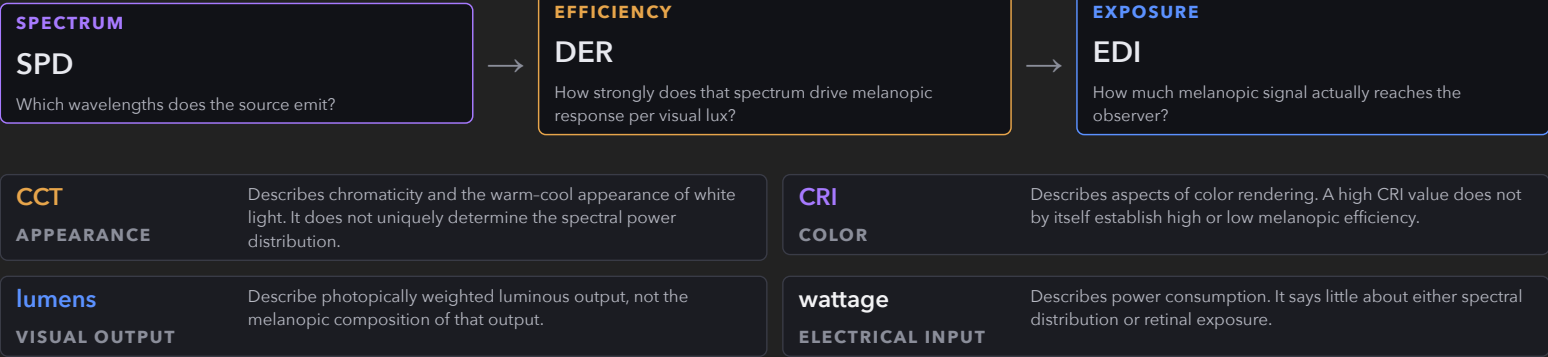
Spectrum → DER → illuminance → EDI → time

DER converts the spectral question into a useful ratio. Photopic illuminance determines how much of that spectral engine reaches the eye. Their combination gives melanopic EDI; timing determines how that exposure should be interpreted.

FROM SOURCE TO EXPOSURE

Keep the quantities separate.

Each quantity answers a different question. Confusion starts when one is used as a substitute for another.



COMMON SHORTCUTS

Useful specifications, different questions.

These values matter. They simply do not replace spectral or melanopic measurements.

THE USEFUL QUESTION

What signal reaches the eye, and when?

Do not reduce a source to “warm,” “cool,” “healthy,” or “circadian.” Read its spectrum, use melanopic DER to understand its spectral efficiency, combine that with illuminance to determine melanopic EDI, and interpret the result within the biological schedule.

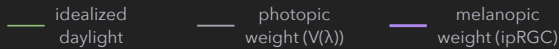
White light has different engines.

Common sources that all appear white do not feed the visual and melanopic readers the same way.

READING THE NUMBER

Melanopic efficiency relative to daylight.

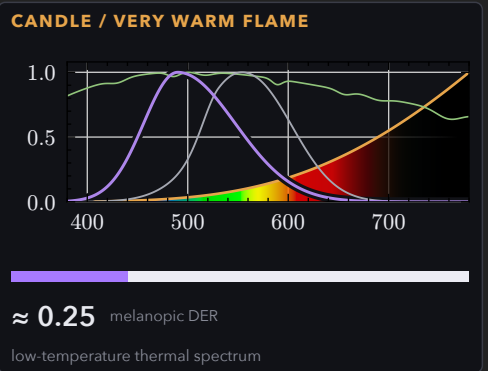
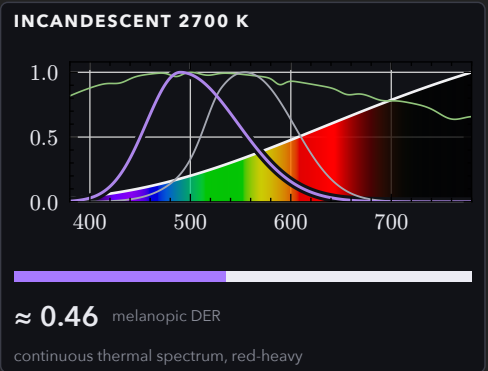
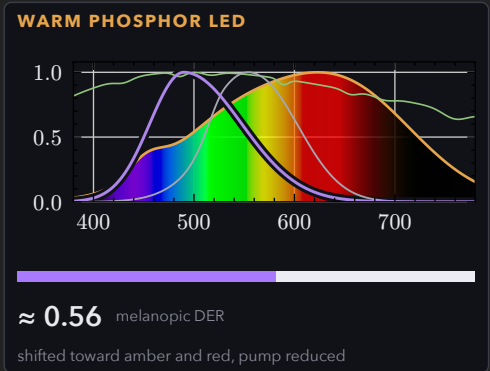
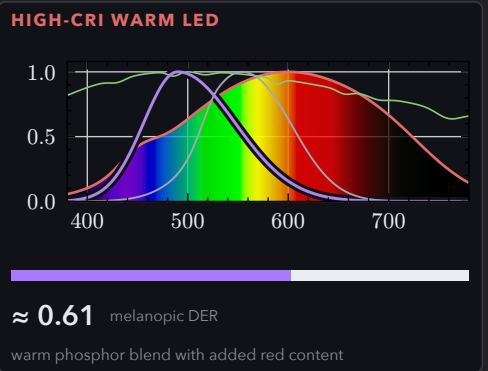
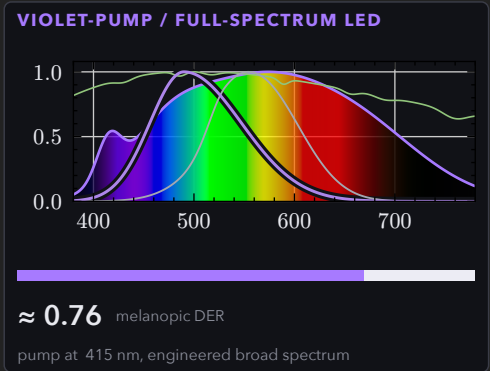
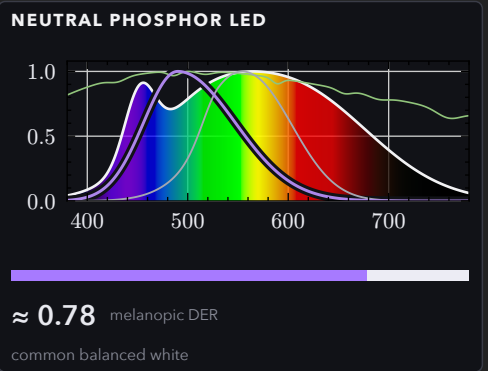
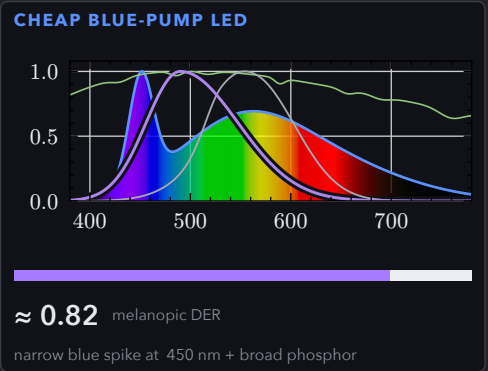
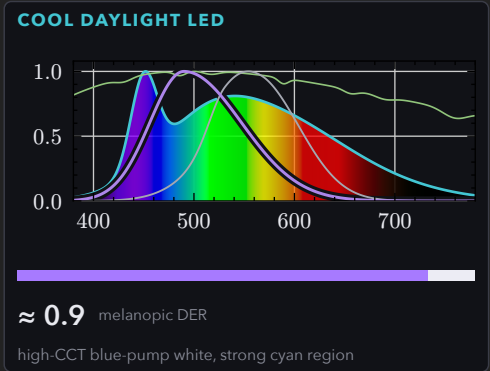
Each modeled source is evaluated by melanopic DER: melanopic response per unit of photopic illumination, referenced to D65 daylight, which has a DER of 1.0. The values are illustrative because the source spectra are representative models rather than measured product spectra.



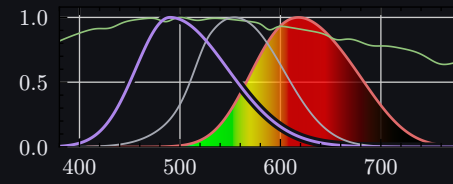
DESIGN CONSEQUENCE

The same visual target can be reached by different spectral engines.

Future interior lighting should specify which engine is active in each scene: broad alerting white by day, warm reduced-blue white in the evening, and narrow long-wavelength light only when biological night must be protected.



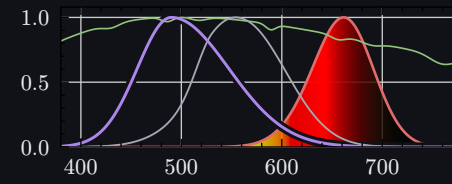
AMBER NIGHT SOURCE



≈ 0.08 melanopic DER

narrow amber/red task and path lighting

DEEP RED NIGHT SOURCE



≈ 0.01 melanopic DER

long-wavelength protection state

Low-blue lamps are a night layer, not a mood.

The night layer should not depend on ordinary warm-white lamps dimmed low. It needs fixtures whose emitted light is spectrally constrained before it enters the room.

LOWER END

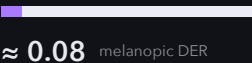
The lower end matters more at night.

The argument is not that every nighttime interior should become red. It is that nighttime scenes need sources whose emitted light avoids the blue-cyan band most strongly read by the circadian system.

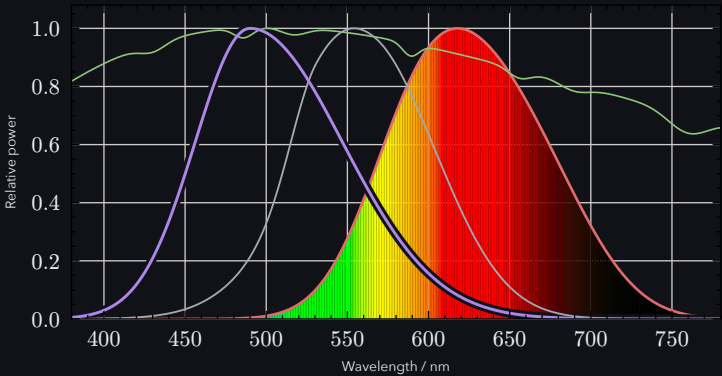
AMBER NIGHT SOURCE

narrow amber/red task and path lighting

MELANOPIC DER, FILL: VIOLET



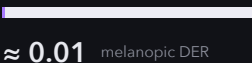
Amber night lighting is not merely warm-looking white. It deliberately avoids the blue-cyan channel and uses the long-wavelength region for orientation, pathfinding, and minimal tasks.



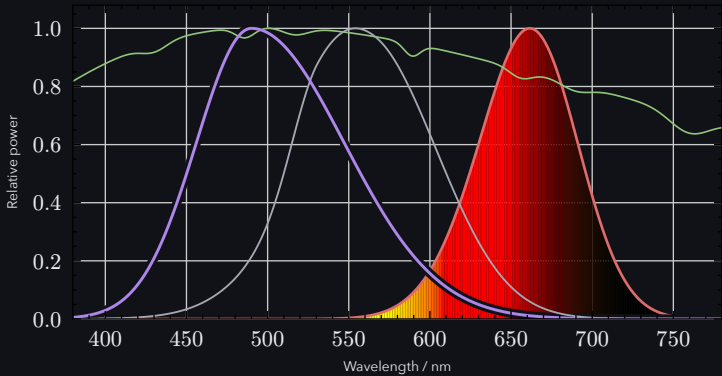
DEEP RED NIGHT SOURCE

long-wavelength protection state

MELANOPIC DER, FILL: VIOLET



Deep red is the extreme case: little useful color rendering, low general visibility, but extremely low melanopic efficiency. It belongs to night protection, not ordinary room lighting.



BLUE-BLOCKING LAMP

The source does the filtering.

A blue-blocking lamp is not eyewear, software, or a habit correction. It is a fixture, bulb, tape channel, or path light whose emitted spectrum removes the blue-cyan band that overlaps the melanopic reader.

NOT WARM-WHITE

Warm appearance is not enough.

A 2200-2700 K lamp can still be a phosphor LED with a blue pump hidden under an amber-looking output. "Warm," "cozy," and "soft white" are visual descriptions; low-blue is a spectral requirement.

WHERE IT BELONGS

Use it in the late-night circulation layer.

The best applications are bedrooms, bathrooms, nurseries, corridors, stair paths, toe-kick lighting, bedside lamps, closet edges, and kitchen night paths: small-area, low-output orientation light.

PRODUCT WARNING

Low-blue does not authorize high output.

A blue-blocking lamp reduces melanopic efficiency; it does not make unlimited light harmless. The target is enough light to move safely without restarting the day signal.

PRODUCT TRANSLATION

For interior lighting, blue blocking means source-side spectral control: lamps and luminaires that avoid the blue-cyan band during biological night. Treat them as a layer below ordinary warm-white lighting – not as daytime white, not as decorative amber ambience, and not as permission to brighten the night.

Good lighting is a function.

The design target is not one optimized white spectrum. It is a day that changes spectral job.

FROM DIMMING

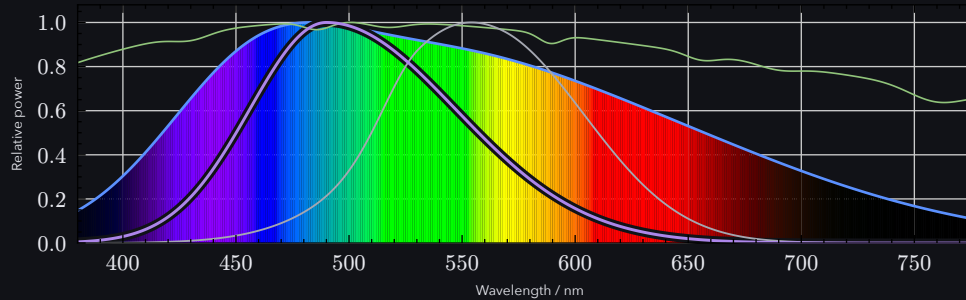
Quantity is not character.

Dimming changes how much light the ipRGC receives. It does not change the spectral efficiency of what remains. A dimmed blue-pump LED is still a blue-pump LED.

STATE 01

Awake Daytime

Drive the ipRGC signal when the body needs it.



VISUAL

High visual output with broad white appearance for work, orientation, and task performance.

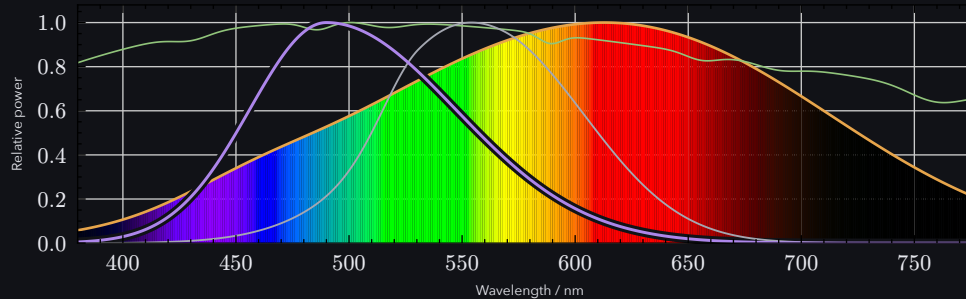
CLOCK

Intentional blue-cyan presence activates melanopsin, suppresses melatonin, and anchors the SCN to daytime.

STATE 02

Evening Transition

Separate visual comfort from circadian activation.



VISUAL

Warm, usable, socially comfortable light – sufficient for faces, tasks, and hospitality.

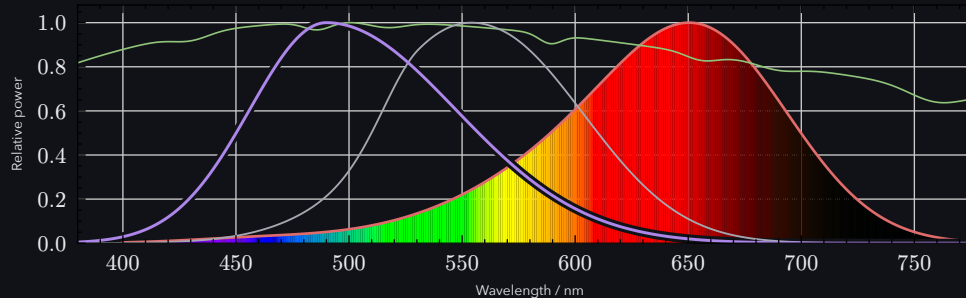
CLOCK

Short-wavelength content is reduced so ipRGC drive falls and melatonin can rise ahead of sleep.

STATE 03

Restful Nighttime

Withdraw the ipRGC signal as completely as possible.



VISUAL

Minimal task light: navigation, safety, and orientation only. Not a room-filling scene.

CLOCK

Blue-cyan energy is minimized as the primary design goal, not as a cosmetic warmth decision.

